

Composite scalable hierarchical layered membrane coupled with passive cooling for sustainable building thermal management

Bencheng Zhao^a, Chen Deng^a, Qiong Tian^a, Tao Zhang^{b,*}, Xuejie Yue^a, Fengxian Qiu^{a,*}

^a School of Chemistry and Chemical Engineering, Jiangsu University, Zhenjiang, Jiangsu Province 212013, China

^b Institute of Green Chemistry and Chemical Technology, Jiangsu University, Zhenjiang, Jiangsu Province 212013, China

ARTICLE INFO

Keywords:

Hierarchical porous structure
Radiative cooling
Fibrous membrane
Building thermal management

ABSTRACT

As global concerns over building energy consumption grow, radiative cooling technology presents a promising solution by providing passive cooling through simultaneous reflection of incident solar irradiance and infrared emission, without the need for external energy input or waste generation. However, most existing radiative cooling materials focus primarily on visible light reflectance and infrared emission, neglecting the near-infrared band, which accounts for nearly half of the total solar energy, which presents a significant opportunity for improvement. In this study, a hybrid electrospun fibrous membrane (LPH) with hierarchical porous structures, composed of polyethylene oxide (PEO) and lithium fluoride (LiF), was fabricated using a PEO/LiF acetonitrile solution. The addition of LiF resulted in a hierarchical microstructure, which significantly enhanced near-infrared reflectance and broadened overall solar reflectance. This novel membrane demonstrates high near-infrared reflectance (90 %) and significantly enhances overall spectral reflectance, contributing to improved radiative cooling performance for building energy savings. With a broadband solar reflectance of 94.22 % and high infrared emissivity of 92.25 %, the LPH membrane achieved a net cooling power of 106 W/m² under direct sunlight (430 W/m²) in validation tests, resulting in a substantial temperature reduction of 9.2 °C, outperforming the pristine membrane by 2.5 °C. These results highlight the exceptional sub-ambient cooling capabilities and the substantial energy-saving potential of this membrane. Moreover, the electrospinning technique facilitates the scalable production of these membranes, making them ideal for large-scale applications in building energy efficiency. This study presents a practical and scalable strategy for optimizing the radiative cooling performance of fibrous membranes, paving the way for sustainable, energy-efficient thermal management solutions in buildings.

1. Introduction

Building energy consumption, constituting over one third of global energy consumption, has raised significant concerns due to its relentless growth trend and the consequential environmental degradation caused by greenhouse gases emission [1–3]. The increasingly frequent extreme weather issues and drastic temperature fluctuations are posing great threats toward human health [4,5]. Confronted with worsening environmental conditions, the prevalent reliance on power-consuming HVAC systems which provide thermal comfort within confined zones could result in a net heating effect and a detrimental cycle, exacerbating rather than alleviating the challenges associated with building energy consumption [6]. Given that building cooling energy consumption takes up a significant portion of total building energy usage, particularly in residential buildings within hot regions and workshops generating

continuous waste heat, reducing cooling consumption emerges as a promising strategy for effectively decreasing overall building energy consumption [7,8]. As a promising and thriving thermal management technology, radiative cooling technology which enables objects to get cooled without any energy input or waste output has garnered significant research attention and experienced prosperous development [9–11]. To be specific, cooling is achieved through strong solar reflectance across the 0.4–2.5 μm wavelength range and enhanced infrared emission, particularly within the atmospheric window (7–14 μm) [12]. This technology shows potential for application in energy-efficient buildings, with the capability to significantly reduce energy consumption and mitigate environmental impact [13,14].

In recent years, numerous radiative cooling materials have been developed and applied across various fields, showcasing a diverse array of structural designs, including porous materials [15–17], metamaterials

* Corresponding authors.

E-mail addresses: zhangtaochem@163.com (T. Zhang), fxqiu@ujs.edu.cn (F. Qiu).

<https://doi.org/10.1016/j.conbuildmat.2025.142593>

Received 21 February 2025; Received in revised form 27 May 2025; Accepted 5 July 2025

Available online 7 July 2025

0950-0618/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

[18,19], photonic crystals [20,21], coatings [22,23], and fibrous membranes [24–28]. Among these, fibrous membranes are particularly promising for radiative cooling due to ease of scalable fabrication and porous structure, which enhances optical performance and supports multifunctional applications [29]. Recent works have demonstrated the effectiveness of electrospun membranes with hierarchical porosity in achieving high solar reflectance and cooling performance. For instance, Wu et al. designed and fabricated a multilayer silk textile via electrospinning [30]. Benefiting from the hierarchical porous structure, the silk textile shows ultrahigh solar reflectance (96.5 %). Coupled with a high infrared emittance of 97.1 %, the textile achieves a notable sub-ambient temperature reduction of 5.1 °C under intense solar radiation. Notably, combining polymers with inorganic fillers in electrospun fibrous membranes has proven especially effective in enhancing both solar reflectance and infrared emissivity. For example, Du et al. developed a PEO/SiO₂ hybrid electrospun membrane with a hierarchically porous structure and strong interfacial binding [31]. The incorporation of SiO₂ not only enhanced the solar reflectance due to increased scattering interfaces but also improved the membrane's infrared emissivity, demonstrating the potential of polymer–inorganic hybrid systems for high-performance radiative cooling.

Compared to conventional radiative cooling materials like metals and ceramics [32], common polymers have advantages in easy processability and high infrared emissivity [33,34]. Fibrous polymer membranes, with numerous interlaced fibers and extensively distributed void structures, not only provide a robust scattering basis but vast refractive interfaces, prominently enhancing the reflectance [35]. According to the Mie scattering theory, light with wavelengths comparable to the fiber diameter or pore size undergoes strong scattering [36]. For fibrous membranes, a uniform fiber diameter or pore size could lead to intense reflection of light within a specific wavelength range. However, for practical radiative cooling applications, it is preferable for the material to exhibit broadband reflectance across the solar spectrum, with particularly high reflectance in the near infrared band occupying half of incident solar energy. Hence, fibrous membranes with a wide distribution of fiber diameters and pore sizes within the wavelength range of sunlight endows greater application potential. This structure, known as a hierarchical microstructure, has already found applications in radiative cooling field [37]. For example, Lin et al. designed a cooling ceramic with hierarchical microstructure inspired by *Cyphochilus* scale, exhibiting near-ideal solar reflectance (99.6 %) due to its polydisperse surface features and internal voids [38]. These findings highlight the importance of hierarchical morphology in enabling efficient solar scattering, and motivate our design of a hybrid fibrous membrane with controlled multi-scale structure [39,40].

In this work, a hybrid electrospun fibrous membrane (LPH) featuring a hierarchical porous structure formed by fibers with a wide diameter distribution was fabricated and utilized for building radiative cooling. By introducing ionic salts (LiF) into the electrospinning solution, the resulting membrane exhibits enhanced NIR reflectance due to both morphological features and potential optical contributions from the inorganic phase. Compared to the PEO membranes with uniform fibers, the LPH membrane achieves a higher NIR reflectance (90 %) and a weighted average solar reflectance of 94.22 %, while also showing great infrared emissivity (92.25 %). Practical field test demonstrated a sub-ambient temperature drop of 8 °C under direct sunlight. This work highlights the role of hierarchical structure design in advancing the optical performance and practical applicability of fibrous membranes for passive radiative cooling in buildings.

2. Experimental section

2.1. Fabrication of hybrid membrane

The hybrid membrane was fabricated via electrospinning technique. Initially, a solution with a mass fraction of 10 % was prepared by

dissolving polyethylene oxide (PEO) powder into acetonitrile, followed by vigorous stirring for 3 h to ensure a homogeneous and transparent solution. Subsequently, lithium fluoride (LiF) powder was introduced into the solution in a gradual and intermittent manner, allowing for the formation of a uniform dispersion. Finally, the electrospinning process was carried out under controlled conditions: a voltage of 20 kV, a spinning distance of 16 cm, and a feeding rate of 0.05 mL/min, utilizing a 22-gauge needle. The collector roller, measuring 30 cm in width and 6.5 cm in diameter, operated at a rolling rate of 120 r.p.m. The thickness of obtained membrane could be adjusted via controlling the spinning solution volume. The obtained LiF/PEO hybrid fibrous membrane (denoted as LPH membrane) was dried naturally before utilization. Pure electrospun fibrous PEO membrane (denoted as PEO membrane) was prepared via the above process without introducing LiF. The preparation procedure is summarized in Scheme 1. The chemicals and characterization could be found in Supporting Information.

2.2. Thermal management performances

To evaluate the thermal management performance of the LPH membranes, two distinct experimental setups were designed to test their response in the solar and far-infrared bands, respectively. A Xenon lamp (CEL-PE300–4A) was used as the light source to simulate solar radiation, while a thermostatic heating stage (IKA C-MAG-HS4) was employed to replicate conditions similar to building exterior walls. For the solar band tests, the Xenon lamp was operated at three different illumination intensities: 0.5, 1, and 1.5 sun (where 1 sun equals 1 kW/m²). Temperature measurements were conducted using thermocouples placed beneath the samples, and the data were recorded with a thermometer (TES-1384). For the far-infrared band tests, the thermostatic heating stage was maintained at a stable temperature of 40 °C. Additionally, an infrared imaging camera (FLIR ONE PRO) was utilized to capture the infrared irradiance output and surface temperatures of the samples. All above-mentioned tests were conducted in a windless indoor environment.

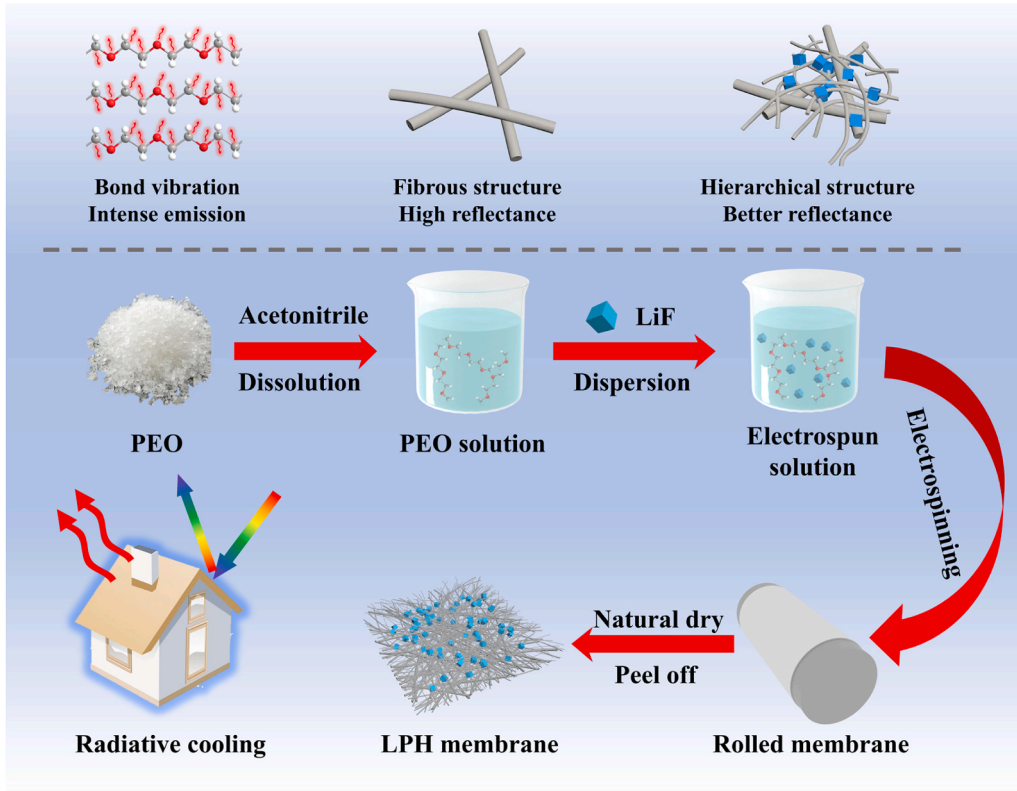
2.3. Validation test of thermal management properties

An experimental setup comprised a thermal insulation foam box encased in solar reflective aluminum foils, along with a chamber room constructed of plastic to accommodate the samples. The LPH membrane and the PEO membrane were trimmed to the same size and placed inside the chamber. Subsequently, the entire setup was relocated onto a shelf fixed on flat ground to stay on a certain height avoiding heat input from the ground. The temperatures of these membranes were measured by two thermocouples inserted underneath the membranes and data logged by the thermometer. The solar irradiance was also measured and recorded by a solar power meter (TES-132). The ambient temperature was measured using a standard Stevenson screen.

2.4. Evaluation of optical properties

The solar reflectance spectra (0.4–2.5 μm) of the PEO and LPH membrane were conducted utilizing a UV–vis–NIR spectrophotometer (Agilent Cary 6000i) equipped with a diffuse integrating sphere. And a BaSO₄ base panel served as the reference across the corresponding wavelength range. The infrared reflectivity and transmission spectrum (5–25 μm) of the LPH membrane were assessed using a Nicolet iS50 spectrophotometer. The weighted average reflectance in the solar spectrum ($\bar{R}_{solar}$) is calculated using Eq. (1). While the average emissivity ($\bar{\epsilon}$) in the infrared band ($\bar{\epsilon}_{IR}$) and the atmospheric window band ($\bar{\epsilon}_{ATW}$) is defined as Eqs. (2 and 3), respectively.

$$\bar{R}_{solar} = \frac{\int_{0.4\mu m}^{2.5\mu m} I_{solar}(\lambda) \cdot R(\lambda) d\lambda}{\int_{0.4\mu m}^{2.5\mu m} I_{solar}(\lambda) d\lambda} \quad (1)$$



Scheme 1. Schematic illustration of the fabrication process and radiative cooling mechanism of the LPH membrane.

$$\bar{\epsilon}_{IR} = \frac{\int_{5\mu m}^{25\mu m} I_{BB}(\lambda, T) \cdot \epsilon(\lambda) d\lambda}{\int_{5\mu m}^{25\mu m} I_{BB}(\lambda, T) d\lambda} \quad (2)$$

$$\bar{\epsilon}_{ATW} = \frac{\int_{7\mu m}^{14\mu m} I_{BB}(\lambda, T) \cdot \epsilon(\lambda) d\lambda}{\int_{7\mu m}^{14\mu m} I_{BB}(\lambda, T) d\lambda} \quad (3)$$

$$I_{BB}(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{(e^{\frac{hc}{\lambda kT}} - 1)} \quad (4)$$

Here, $R(\lambda)$ is the spectral reflectance in band from 0.4 to 2.5 μm , $I_{solar}(\lambda)$ is the standard solar spectrum characterizing the spectral distribution of solar radiation arriving at the surface. In Eqs. (2) and (3), $\epsilon(\lambda) = 1 - \rho(\lambda) - \tau(\lambda)$ is the spectral infrared emissivity. $I_{BB}(\lambda, T)$ is the spectral radiance of the blackbody at T . In Eq. (4), h , c and k are the Planck constant (6.626×10^{-34} J·s), the speed of light (3×10^8 m/s) and Boltzmann constant (1.381×10^{-23} J/K), respectively.

3. The theoretical model of LPH radiative cooling membrane performance

To evaluate the cooling performance of the LPH membrane, a direct thermal measurement system was built, and the corresponding schematic diagram is presented in Fig. S1A [12]. When the LPH membrane is exposed to a clear sky in an open environment, its surface radiates thermal energy to the sky while simultaneously absorbing both solar irradiance and the downward thermal radiation emitted by the atmosphere. Additionally, heat transfer from the surrounding environment to the membrane occurs via conduction and convection, driven by the temperature difference between it and the ambience. Based on the above analysis, a theoretical model of radiative cooling can be established, considering the infrared radiation power of the membrane, the absorbed solar radiation, the convection and thermal conductivity with the surrounding environment, and the radiative reverse heat flow from the

atmosphere. The net cooling power density is thus expressed as Eq. (5):

$$P_{net} = P_{rad} - P_{solar} - P_{atm} - P_{conv} - P_{set} \quad (5)$$

where P_{rad} is the power density of thermal radiation emitted by the LPH membrane;

P_{atm} is the power density of the downward thermal radiation emitted by the ambient atmosphere;

P_{solar} is the heating power density of incident solar irradiance adsorption;

P_{conv} is the convective and conductive power density from the top surface of the LPH membrane;

P_{set} is the thermal leakage input from the thermal isolation experimental setup.

P_{rad} and P_{atm} are determined by both the spectral data of the LPH membrane and the emissivity spectrum of the atmosphere.

The power density of the absorbed solar irradiance can be calculated by Eq. (6):

$$P_{solar} = \int_0^\infty \alpha_{LPH}(\lambda) I_{solar}(\lambda) d\lambda \quad (6)$$

where $I_{solar}(\lambda)$ is the solar spectral irradiance and $\alpha_{LPH}(\lambda)$ is the wavelength-dependent solar absorbance of the LPH membrane. P_{conv} is contributed to by heat conduction and convection from the top surface of the LPH membrane. To minimize conductive and convective heat exchange between the sample and the ambience, a thin layer of polyethylene (PE) film was applied over the thermal isolation box. Additionally, P_{set} represents the thermal leakage from the insulation box.

The thermal radiation power P_{rad} is the only outgoing heat flux enabling the cooling of the LPH membrane. When the membrane's temperature is below ambient, all other power densities (P_{atm} , P_{solar} , P_{conv} and P_{set}) act as inward heat fluxes, opposing the cooling process. To further assess the total radiative cooling power, a polyimide heater was integrated into the setup, providing an additional degree of freedom for

evaluation.

The heat exchange coefficient h_{conv} typically ranges from 5 to 20 W/m²·K. To characterize P_{conv} and P_{set} , experimental methods were employed. As shown in Fig. S1B, a heat flux is fed into the thermal box using the heater to maintain a certain temperature difference between the membrane and ambient surroundings. Throughout this experiment, all radiative heat transfer pathways were blocked, and tests were conducted under indoor conditions. Based on this experimental setup, the h_{conv} is thus calculated to be ~ 5.02 W/m²·K and the leakage rate is ~ 2.94 W/m²·K.

Using the spectral data of LPH membrane, P_{rad} can be evaluated by integrating the infrared emissivity spectrum and the blackbody irradiance. The net cooling power P_{net} was calculated based on the validation test results, conducted under an ambient temperature of 320 K and a solar irradiance of 430 W/m². Also, to study the effects on sub-ambient cooling temperature, various non-radiative heat exchange coefficients from 1.0 W/(m²·K) to 20.0 W/(m²·K) are considered. When the surface is cooled below the ambient, heat transfer from the surrounding environment to the membrane surface occurs through convection and conduction. Meanwhile, P_{rad} decreases as the surface temperature decreases. Eventually, when the thermal radiation power P_{rad} is cancelled out by the other inward heat fluxes, the net cooling power P_{net} declines to zero and the surface temperature reaches steady.

4. Results and discussion

4.1. The morphological and constituent analysis

The crystal construction of obtained LPH membrane was evaluated via XRD test. As shown in Fig. 1A, the peaks located at 19.1°, 23.26° and

26.1° are corresponded to the characteristic peaks of poly (ethylene glycol) (PDF# 00-053-1860). While LiF could be clearly be recognized by the characteristic peaks located at 38.7°, 45.02°, 65.48° and 78.68°, corresponding to the (1 1 1), (2 0 0), (2 2 0) and (3 1 1) planes, respectively (PDF# 01-072-1538). The characteristic peaks are labeled with different tags according to the different substances. These peaks indicate the complete hybrid of PEO and LiF, which is consistent with the EDS analysis (Fig. 1B). The accompanying EDS spectrum indicates the presence of these elements, affirming the uniform distribution and hybrid of LiF within the PEO fibers (Fig. 1(C1-C4)).

The thermogravimetric analysis and differential scanning calorimetry of the LPH membrane are illustrated in Fig. 2. The weight of the LPH membrane remained stable below 200 °C, indicating its adequate thermal stability for practical applications. A minor endothermic peak observed at 77 °C corresponds to the melting phase transition of crystalline PEO. As the temperature increases to approximately 260 °C, a significant weight loss is observed, and the DSC signal turns positive, indicating the start of thermal decomposition of PEO accompanied by heat release. The major thermal decomposition occurred at approximately 400 °C, marked by a rapid increase in the DSC signal and a sharp exothermic peak. After the thorough thermal composition of PEO, the residual weight could be attributed to inorganic component LiF due to its high melting point, revealing a weight of about 27 % of LiF.

Scanning electron microscopy (SEM) was utilized to determine the morphological features of obtained PEO and LPH membrane. In general, the SEM images of both two membranes reveal a fibrous network with a highly porous structure resulting from the electrospun preparation process (Fig. 3A). As shown in Fig. 3(B and C), the fibers appear smooth and well-aligned. Except for a small portion of fibers up to 2 microns in diameter, most fibers have diameters concentrated around 1 micron.

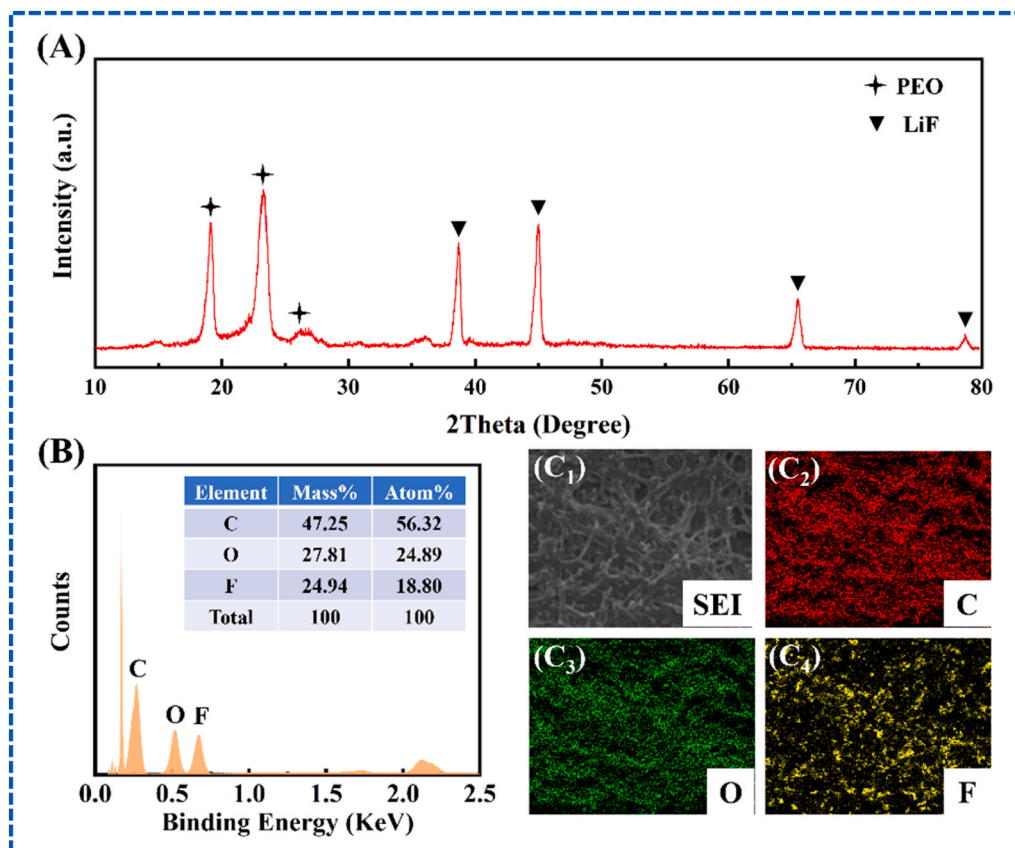


Fig. 1. Structural and compositional characterization of the LPH membrane. (A) XRD pattern of the LPH membrane: peaks marked with “+” correspond to PEO (PDF#00-053-1860), and those marked with “▼” correspond to LiF (PDF#01-072-1538). (B) EDS spectrum and elemental analysis of the LPH membrane. (C₁) SEM image of the LPH membrane. (C₂-C₄) Elemental mapping of carbon (C), oxygen (O), and fluorine (F), respectively.

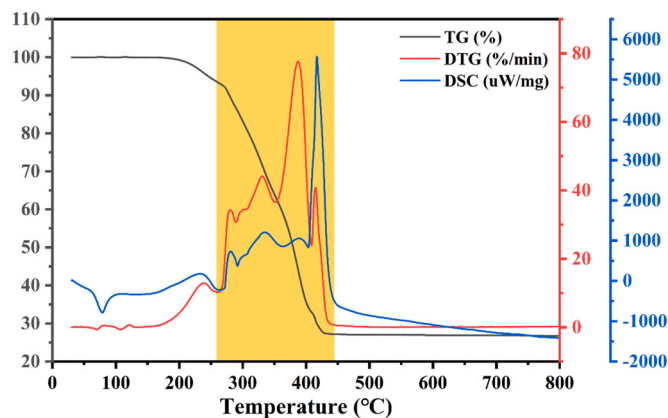


Fig. 2. Thermal analysis of the LPH membrane using TG, DTG, and DSC techniques. The black curve shows the thermogravimetric (TG) profile. The red and blue curves represent the differential thermogravimetric (DTG) and differential scanning calorimetry (DSC) data, respectively. The highlighted temperature range ($\sim 300\text{--}480\text{ }^{\circ}\text{C}$) corresponds to major thermal decomposition events.

The fibrous porous structure with ample void spaces, consisting of numerous fibers with diameters close to the average wavelength of incident sunlight, could strengthen the scattering performance and contribute to high solar reflectance of the PEO membrane. In Fig. 3(D-

G), an appreciable number of particles could be distinguished due to the introduction of LiF powder, which is consistent with the results of EDS. Despite the fibrous and porous network of LPH membrane, several thick fibers with diameters exceeding 2 microns were observed. Also, the interstitial spaces between these fibers were filled with finer nanofibers whose diameters range from tens to hundreds of nanometers exhibiting noticeable branching and entanglement. Correspondingly, this hierarchical porous structure, characterized by the finer nanofibers, leads to a reduced pore diameter distribution of LPH membrane (Fig. S2). The above-mentioned differences affect the morphology features of LPH membrane across multiple scales, broadening the distribution range of fiber diameter and, consequently, altering the light scattering spectrum. Combined with the inherent high solar reflectance and infrared emissivity of LiF, the LPH membrane could enhance both NIR reflectance and IR emission compared to the PEO membrane.

The broad fiber diameter distribution and hierarchical porous morphology of the electrospun membrane are primarily attributed to the electrical conductivity gradient induced by LiF dispersion and dynamic instability during the electrospinning process. The incorporation of LiF increases the solution conductivity, intensifying charge repulsion within the electrified jet, which promotes whipping instability and uneven stretching, leading to varied fiber diameters. Simultaneously, fluctuations in electric field strength and solvent evaporation rates exacerbate jet splitting and non-uniform solidification. Additionally, the extended flight path (e.g., collector distance) allows gradient phase separation between PEO and LiF, coupled with layered fiber deposition, creating interconnected pores and stratified voids. These combined effects from

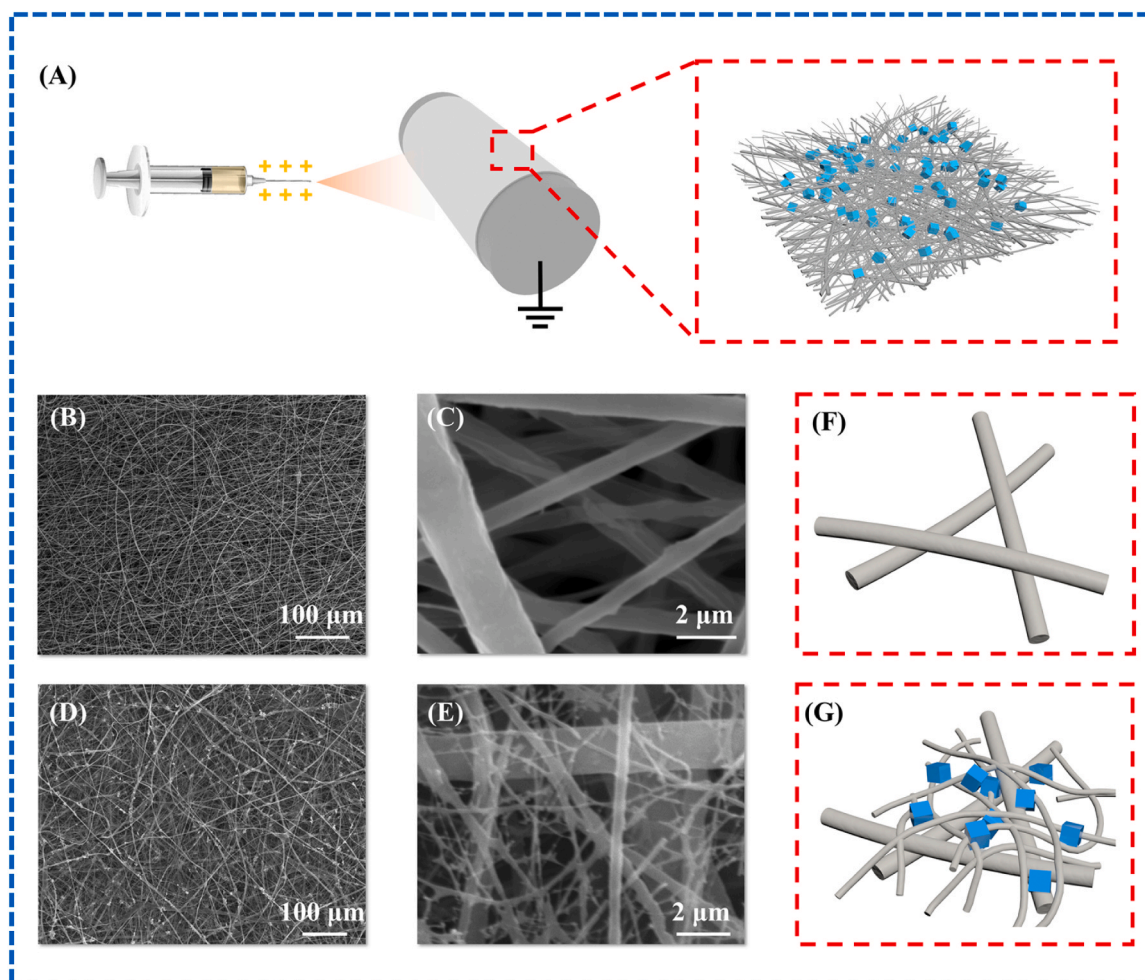


Fig. 3. The schematic of fabrication process of fibrous membranes (A). SEM images of PEO electrospun fibrous membrane (B-C) and LPH membrane (D-E) with different magnification. The schematic of corresponding PEO and LPH membrane's porous structure (F-G).

conductivity heterogeneity and process dynamics result in structurally diverse fibers and multi-scale porosity.

4.2. The optical property analysis

Given the importance of optical properties in regulating the heat input and dissipation process of the building, the solar reflectance and infrared emissivity spectra of the obtained membranes were measured. The reflectance spectra presented in Fig. 4A highlight the high solar reflectance properties of both the PEO and LPH membranes, with weighed average reflectance values of 90.36 % and 94.22 %, respectively. Both membranes exhibit peak reflectance in the visible band, where solar irradiance is most intense, thereby effectively mitigating incident photothermal conversion. The LPH membrane exhibits superior solar reflectance compared to the PEO membrane, particularly in the near-infrared band, indicating intensified suppression of heat absorption by the LPH membrane. This enhancement can be attributed to the additional fibers of varying diameters distributed across the solar radiation wavelengths and the introduction of LiF with high refractive index, which improve the broadband reflectance of the LPH membrane. Also, the hierarchical porous morphology promotes multiple scattering and increases the optical path length, further enhancing reflectance in the NIR region. As shown in Fig. 4B, fibers with specific diameters exhibit distinct scattering peaks at different wavelengths (dash lines). This suggests that the reflection spectrum of a membrane containing fibers with a broad distribution of diameters will exhibit a wider spectral range. The scattering efficiency spectrum of the LPH membrane (black solid line), calculated from the fiber diameter histogram data, demonstrates strong scattering across the entire solar spectrum, further confirming its broadband reflectance. According to the fiber diameter histograms shown in Fig. 4(C and D), the fiber diameter distributions of the two fibrous membranes differ, with the mean fiber diameters of the PEO and LPH membranes being 1583 nm and 754 nm, respectively. This difference can be mainly ascribed to the fiber structure, where the LPH membrane features pronounced branching and finer nanofibers, resulting in a significantly lower mean fiber diameter compared to the PEO

membrane (Fig. S3). The higher proportion of fibers with diameters spanning the visible wavelength range could endow LPH membrane with enhanced visible reflectance performance. Additionally, the corresponding scattering fields of fibers with mean diameters are presented in Fig. 4(E and F), further illustrating the scattering performance differences between PEO and LPH membrane.

As depicted in Fig. 5A, the LPH membrane demonstrates an average infrared emissivity of 92.25 %, with an emissivity of 90.12 % in the atmospheric window band (7–14 μm). This high emissivity facilitates efficient heat dissipation within the infrared spectrum, implying the membrane's radiative cooling potential. To investigate the molecular structure and chemical composition, FTIR spectroscopy was conducted for both PEO and LPH membranes (Fig. 5B). Both spectra exhibit a distinct characteristic absorption peak at 2588 cm^{-1} (marked by blue transparent rectangle), corresponding to the asymmetric and symmetric stretching vibrations of the C-H bonds from PEO methylene groups ($-\text{CH}_2$). In the yellow highlighted region, characteristic peaks for C-H bond bending vibrations are observed at 1466 cm^{-1} and 1342 cm^{-1} . The multiple peaks in the range of 1300–1000 cm^{-1} originate from the vibrations of C-O bonds. Among these peaks, the most intense absorption peak (orange rectangle) at 1098 cm^{-1} is attributed to the C-O-C asymmetric stretching vibrations. Visibly, the aforementioned vibration peaks of C-H and C-O bonds, as illustrated in Fig. 5C, are located within the atmospheric window band, which corresponds to the wavenumber range of 1430–714 cm^{-1} . These vibrations contribute significantly to the membrane's high emissivity in this spectral region. Moreover, the lattice vibrations belonging to LiF results in a broad absorption peak in the fingerprint region. However, it exerts minimal impact on the molecular structure of PEO, as evidenced by the retention of characteristic absorption peaks. The emissivity data and FTIR analysis collectively demonstrate the chemical composition and structural integrity of the LPH membrane. These factors contribute to its superior broadband infrared emissivity, further supporting its potential for radiative cooling applications.

By comparing several key factors of current radiative cooling fibrous membranes (as shown in Table S1), the emphasized unique

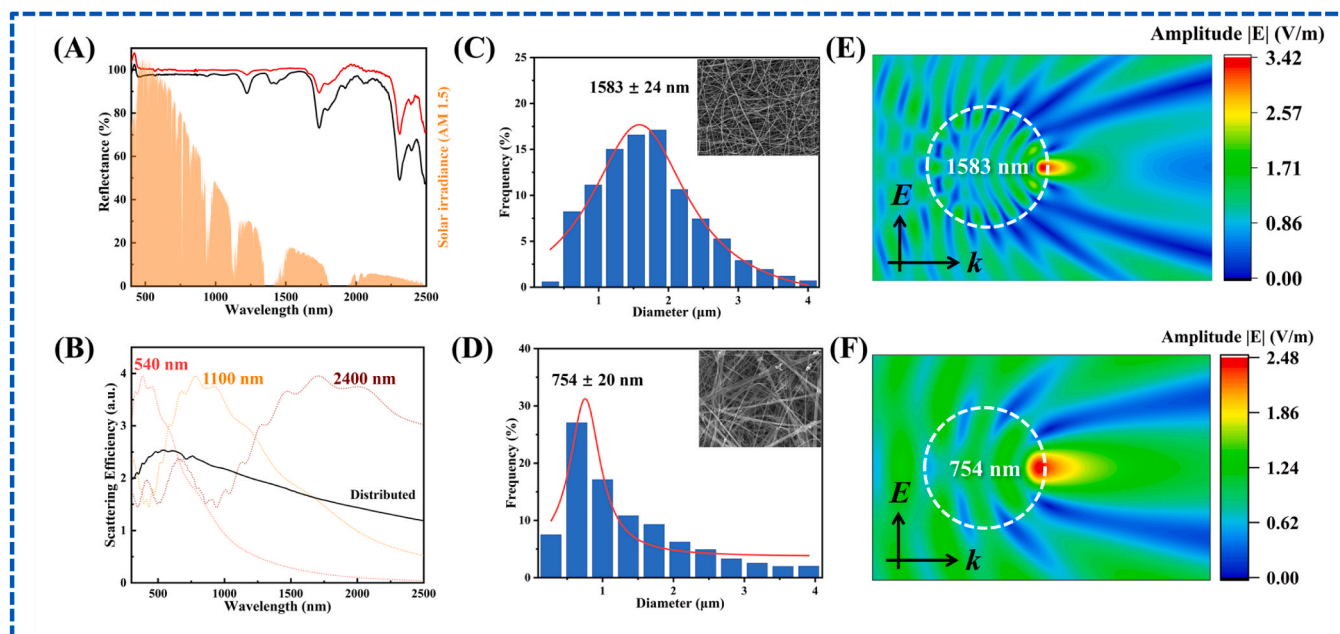


Fig. 4. (A) The visible-near infrared band reflectance spectra of pure PEO electrospun fibrous membrane (black) and LPH membrane (red), with AM1.5 solar irradiance (orange area). (B) Calculated scattering efficiency of the hierarchically porous structure of LPH membrane, including the scattering efficiencies of the three primary fibers individually, and the overall membrane scattering efficiency derived from the weighted calculation. (C, D) Fiber diameter histograms and mean diameters of the pure PEO membrane and the LPH membrane, respectively. The insets show the corresponding morphologies. (E, F) Calculated scattering fields of 632-nm incident light by fibers of two mean diameters. E , electric field of the incident light; k , wave vector of the incident light.

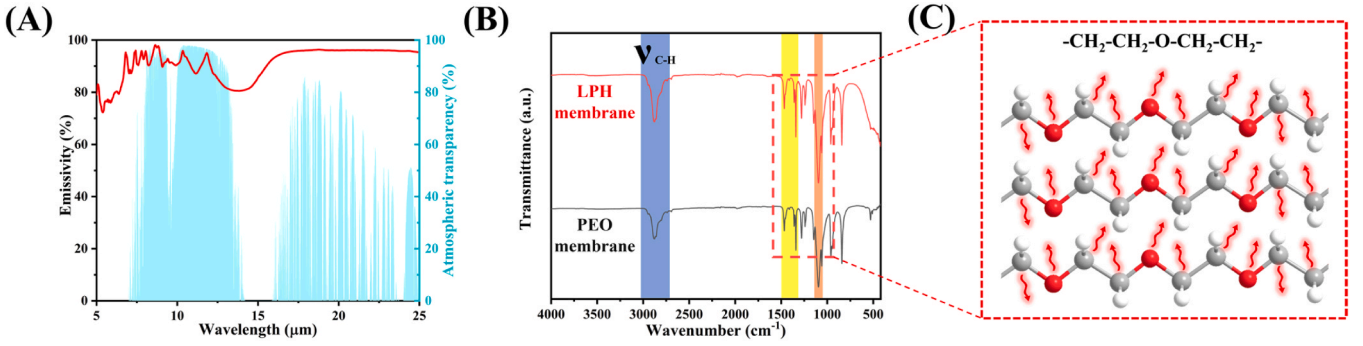


Fig. 5. (A) Emissivity of LPH membranes across the wavelength range of 5–25 μm, overlaid with the atmospheric transparency window (cyan area). (B) The FTIR spectrum of the PEO and LPH membrane. (C) The schematically illustration of the strong infrared adsorption by bond vibrations of PEO molecular chain.

microstructure of the LPH membranes contributes to enhanced radiative cooling performance. In addition, electrospinning, as a scalable approach for developing advanced radiative cooling materials, demonstrates its potential for improving thermal management and energy savings in sustainable building applications.

4.3. The determination and validation tests

To demonstrate the optical properties of the PEO and LPH membranes, two distinct experimental setups were established to simulate the solar irradiation and emissivity-driven heat dissipation conditions, respectively. The test results of simulated solar irradiation are presented in Fig. 6A. As the irradiance intensity increases, the terminal temperatures of both membranes rise accordingly, from 26.3 °C to 42.4 °C for the PEO membrane and 25.2 °C to 40.2 °C for the LPH membrane. Notably,

temperatures of the PEO membrane are consistently higher than that of the LPH membrane under different irradiance levels, demonstrating the better heat input depression performance of the LPH membrane stem from the higher solar reflectance. While for the infrared band, both membranes were placed on a thermostatic heating stage set at 30 °C and imaged via an infrared camera to capture the irradiance outputs and surface temperatures. Despite the hybrid of LiF, the two membranes exhibited similar infrared emissivity, as they share the same base material. The LPH membrane showed only a slight 0.2 °C temperature reduction compared to the PEO membrane, which can be observed in Fig. 6B.

The enhanced solar reflectance and comparable infrared emissivity of the LPH membrane demonstrate the synergistic interplay of heat input suppression and output enhancement and its potential for radiative cooling applications. Furthermore, to validate the radiative cooling

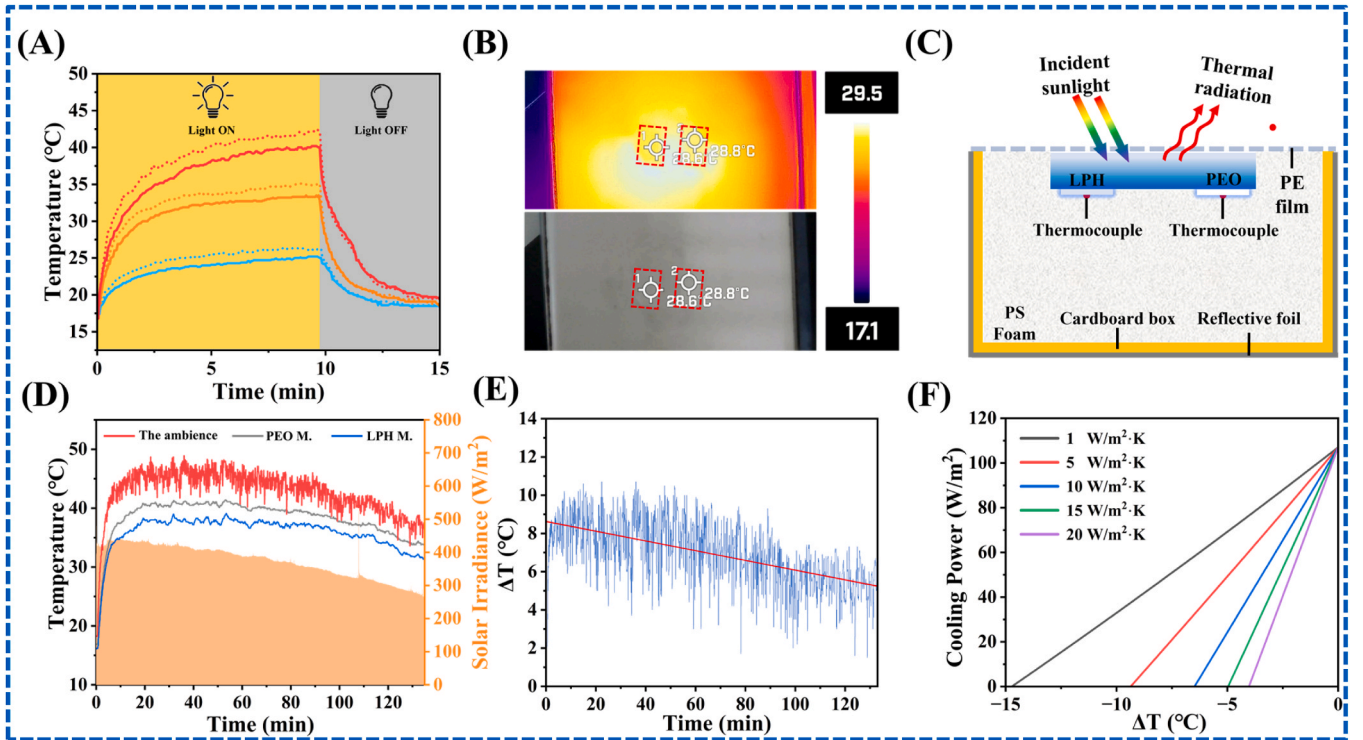


Fig. 6. Temperature measurements of PEO and LPH membranes (dash-dot and solid curves, respectively) under simulated solar irradiation with varying irradiance power levels (blue-0.5 sun, orange-1 sun, red-1.5 sun) during the solar reflectance performance test. (B) Photos and infrared images of PEO and LPH membranes (denoted as 1 and 2 in images, respectively). The membranes are indicated by dashed-line boxes. (C) Schematic diagram of the experimental setup used for the feasibility test. (D) Temperature measurements in the feasibility test under corresponding solar irradiance (orange area). (E) Temperature differences between the ambient and the LPH membrane during the test. (F) the radiative cooling power of the LPH membrane versus sub-ambient cooling temperature for various non-radiative heat exchange coefficients. The calculations are based on the actual spectral data of the LPH membrane.

performance, an experimental setup was constructed using a cardboard box covered with a reflective film (Fig. 6C). Thermal-insulation polystyrene foam was placed inside the box to prevent external heat invasion. A cavity of several millimeters deep was carved into the foam, minimizing the cavity's volume and consequent chamber heating during the test. And the cavity was sealed with a PE film to reduce the influence of environmental parasitic heat. Three thermocouples were used to monitor temperature changes over time, with two placed beneath the PEO and LPH membranes and one in the ambient environment.

The validation test was conducted on a clear, breezy day in Zhenjiang, Jiangsu, on January 12, 2024. Specifically, the experimental apparatus was exposed to solar irradiance at noon for over two hours, a period characterized by intense sunlight and a high solar illumination angle, ensuring strong direct irradiance. As shown in Fig. 6D, under a solar irradiance of approximately 430 W/m^2 , the temperatures of the PEO membrane, LPH membrane, and the ambience rise rapidly, peaking at 40.8°C , 38.3°C , and 47.5°C , respectively. Both membranes exhibit sub-ambient cooling, confirming the effective radiative cooling performance of PEO-based fibrous membranes. Notably, the LPH membrane exhibits superior cooling performance with an additional temperature reduction of 2.5°C because of the hierarchically porous microstructure and incorporation of LiF. During the remaining period of irradiance, all temperatures decrease with the slow attenuation of irradiance power. Also, temperature differences get narrowed. While, the temperature drops between the LPH membrane and the ambience consistently maintains 6.7°C in average, highlighting its sustained cooling efficiency (Fig. 6E). Beside the cooling efficiency, the practical applicability of the LPH membrane was assessed via an additional test under summer conditions with intensified and fluctuating solar radiation using the same batch of aged samples. The results are provided in Fig. S4. During this test, the LPH membrane still maintained lower temperature than both the PEO membrane and the ambience. The accompanying variations relative humidity also supports the membrane's adaptability to dynamic outdoor conditions.

To further evaluate the radiative cooling potential of the LPH membrane, the theoretical model of radiative cooling performance was

established. Under the test conditions, the calculated net radiative cooling power of the LPH membrane reaches 106 W/m^2 , exhibiting a competitive radiative cooling performance. To evaluate the effects of heat conduction and convection on the sub-ambient cooling temperature, various non-radiative heat exchange coefficients are considered and the corresponding curves are shown in Fig. 6F. The crossing points of the curves on the ΔT axis are the lowest temperatures that could be cooled by the LPH membrane. at the condition with $h_{\text{conv}} = 5 \text{ W/(m}^2\text{K)}$ and an ambient temperature of 320 K , the membrane could provide a temperature drop of 9°C . When h_{conv} declines to $1 \text{ W/(m}^2\text{K)}$, the sub-ambient temperature could raise to around 15°C .

4.4. Energy saving estimation

To estimate the energy saving potential of the obtained LPH membrane in practice, a building thermal load simulative calculation software DeST was utilized to calculate the energy consumption (the parameters of building model can be accessed in Table S2). As depicted in Fig. 7(A-C), due to the competitive radiative cooling performance of the LPH membrane, the building heat load in summer days decreased with membrane attached. Detailly, the cooling load decreases by 18 %, 17 % and 14 % for Beijing, Shanghai and Haikou, respectively. Although the radiative cooling performance brings overcooling in cold weather, the overall annual energy consumption maintains declined, implying the huge building energy saving potential of the obtained membranes (Fig. 7D). Based on the above results, it can be supposed that the material will exhibit superior cooling and energy-saving performance in regions with hot climates.

5. Conclusion

In conclusion, an electrospun hybrid fibrous LPH membrane exhibiting high solar reflectance (94.22 %) and comparable infrared emissivity (92.25 %) was fabricated via a facile and scalable strategy. The introduction of LiF powder leads to the formation of hierarchically porous structure characterized by varying fiber diameters and finer

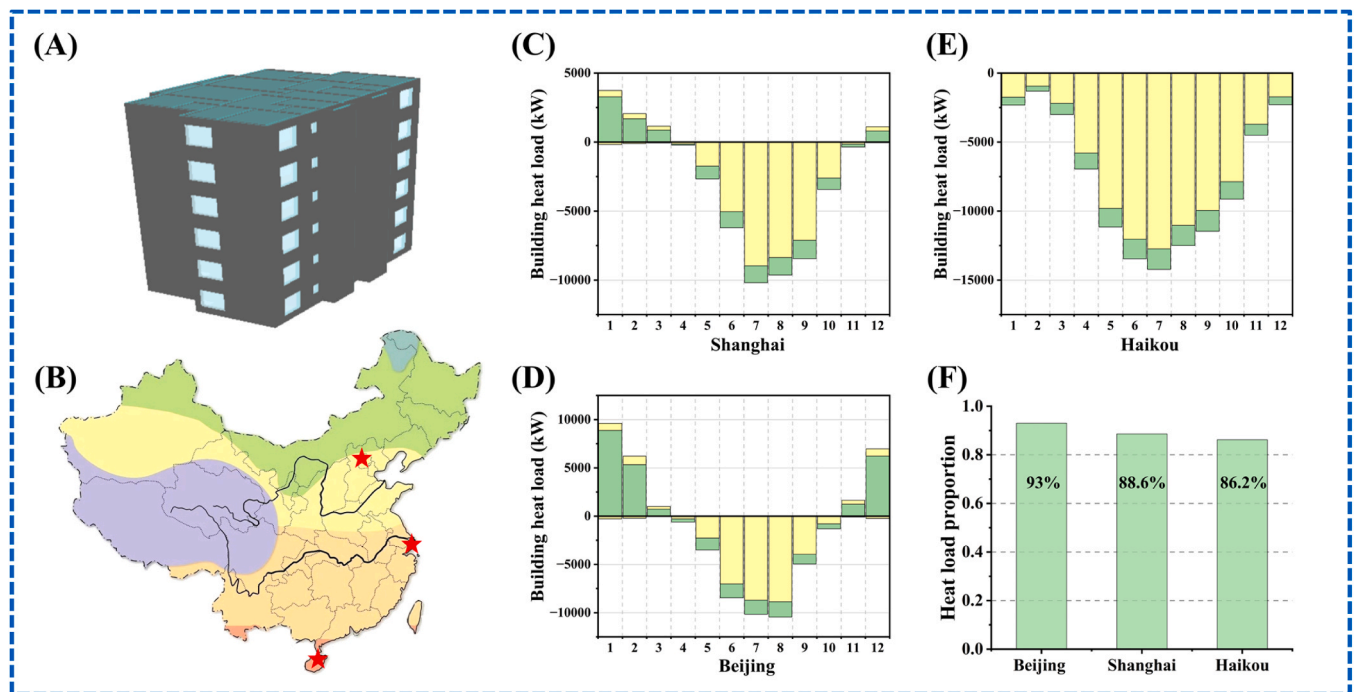


Fig. 7. The building model structured for heat load simulative calculation (A). Map of China by climate, with the three selected cities marked with stars (B). Comparison of building heat loads with and without the LPH membrane in selected cities based on local climate conditions (C-E), and annual thermal energy consumption of buildings with the LPH membrane compared to pristine ones (F).

nanofibers filling the interstitial spaces between larger fibers. And the wide distribution of fiber diameters broadens the scattering spectrum of incident light, contributing to the obtained membrane's superior near infrared reflectance and resulting broadband solar reflectance spectrum. Moreover, the inherent optical properties of LiF, such as its high refractive index and scattering capability, synergistically improve the solar reflectance. Consequently, the weighted solar reflectance of the LPH membrane is increased by approximately 4 % compared to the PEO membrane, marking a substantial improvement in its optical performance. Due to the desirable optical properties, the LPH membrane could offer a remarkable net radiative cooling power of 106 W/m² and temperature drop of 9.2 °C under direct sunlight of 430 W/m², surpassing the PEO membrane by 2.5 °C. Additionally, the energy consumption estimation shows that the buildings loaded with LPH membranes have good energy-saving effects in different cities, and the saving efficiencies averages around 12 %. Moreover, the more hot days per year in the region, the more significant the energy saving effect. The simulation demonstrates the huge potential in practical building energy saving. This work not only highlights the unique microstructure of LPH membranes in achieving energy-efficient cooling but also establishes a scalable approach for developing advanced radiative cooling materials for sustainable building energy saving applications.

CRediT authorship contribution statement

Fengxian Qiu: Project administration, Resources, Supervision, Funding acquisition. **Xuejie Yue:** Formal analysis, Visualization. **Chen Deng:** Investigation, Methodology, Conceptualization. **Tao Zhang:** Methodology, Resources, Writing – review & editing. **Qiong Tian:** Software, Visualization, Formal analysis. **Bencheng Zhao:** Writing – original draft, Data curation, Writing – review & editing, Investigation, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (22408129), Nature Science Foundation of Jiangsu Province (BK20240865), China Postdoctoral Science Foundation (2023M731355) and Graduate scientific research innovation program of Jiangsu Province (KYCX24_4010).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.conbuildmat.2025.142593](https://doi.org/10.1016/j.conbuildmat.2025.142593).

Data availability

Data will be made available on request.

References

- [1] X. Yin, R. Yang, G. Tan, S. Fan, Terrestrial radiative cooling: using the cold universe as a renewable and sustainable energy source, *Science* 370 (6518) (2020) 786–791.
- [2] J. Gao, X. Zhong, W. Cai, H. Ren, T. Huo, X. Wang, Z. Mi, Dilution effect of the building area on energy intensity in urban residential buildings, *Nat. Commun.* 10 (1) (2019) 4944.
- [3] Y. Chen, J. Ran, X. Yue, D. Yang, T. Zhang, F. Qiu, Cellulose-based thermal management composites: an emerging material for multi-industry sustainability, *Renew. Sustain. Energy Rev.* 218 (2025) 115801.
- [4] Q. Zhang, S. Wang, X. Wang, Y. Jiang, J. Li, W. Xu, B. Zhu, J. Zhu, Recent progress in daytime radiative cooling: advanced material designs and applications, *Small Methods* 6 (4) (2022) 2101379.
- [5] Y. Peng, W. Li, B. Liu, W. Jin, J. Schaadt, J. Tang, G. Zhou, G. Wang, J. Zhou, C. Zhang, Y. Zhu, W. Huang, T. Wu, K.E. Goodson, C. Dames, R. Prasher, S. Fan, Y. Cui, Integrated cooling (i-cool) textile of heat conduction and sweat transportation for personal perspiration management, *Nat. Commun.* 12 (1) (2021) 6122.
- [6] X. Li, J. Peoples, Z. Huang, Z. Zhao, J. Qiu, X. Ruan, Full daytime sub-ambient radiative cooling in commercial-like paints with high figure of merit, *Cell Rep. Phys. Sci.* 1 (10) (2020) 100221.
- [7] J. Yuan, H. Yin, D. Yuan, Y. Yang, S. Xu, On daytime radiative cooling using spectrally selective metamaterial based building envelopes, *Energy* 242 (2022) 122779.
- [8] P. Li, Y. Liu, X. Liu, A. Wang, W. Liu, N. Yi, Q. Kang, M. He, Z. Pei, J. Chen, P. Jiang, W. Li, H. Bao, X. Huang, Reversed yolk-shell dielectric scatterers for advanced radiative cooling, *Adv. Funct. Mater.* 34 (2024) 2315658.
- [9] H. Zhai, D. Fan, Q. Li, Dynamic radiation regulations for thermal comfort, *Nano Energy* 100 (2022) 107435.
- [10] S. Fan, W. Li, Photonics and thermodynamics concepts in radiative cooling, *Nat. Photonics* 16 (3) (2022) 182–190.
- [11] Y. Chen, H. Chen, Y. Dai, J. Xiao, F. Qiu, T. Zhang, Honeycomb porous regenerated cellulose aerogel films with enhanced thermal dissipation for agricultural mulch application, *Food Bioprod. Process.* 147 (2024) 418–427.
- [12] J. Liu, H. Tang, C. Jiang, S. Wu, L. Ye, D. Zhao, Z. Zhou, Micro-nano porous structure for efficient daytime radiative sky cooling, *Adv. Funct. Mater.* 32 (2022) 2206962.
- [13] J. Mandal, Y. Yang, N. Yu, A.P. Raman, Paints as a scalable and effective radiative cooling technology for buildings, *Joule* 4 (7) (2020) 1350–1356.
- [14] T. Li, Y. Zhai, S. He, W. Gan, Z. Wei, M. Heidarinejad, D. Dalgo, R. Mi, X. Zhao, J. Song, J. Dai, C. Chen, A. Aili, A. Velloro, A. Martini, R. Yang, J. Srebric, X. Yin, L. Hu, A radiative cooling structural material, *Science* 364 (6442) (2019) 760–763.
- [15] B. Zhao, X. Yue, Q. Tian, F. Qiu, Y. Li, T. Zhang, Bio-inspired bc aerogel/pva hydrogel bilayer gel for enhanced daytime sub-ambient building cooling, *Cellulose* 29 (14) (2022) 7775–7787.
- [16] B. Xiang, R. Zhang, Y. Luo, S. Zhang, L. Xu, H. Min, S. Tang, X. Meng, 3d porous polymer film with designed pore architecture and auto-deposited sio₂ for highly efficient passive radiative cooling, *Nano Energy* 81 (2021) 105600.
- [17] S. Huang, M. Wang, Y. Dai, C. Deng, S. Xue, F. Qiu, T. Zhang, Waste to treasure: Upcycling waste express packing to sustainable packaging materials, *J. Environ. Chem. Eng.* 13 (1) (2025) 115016.
- [18] J. Brewer, S. Kulkarni, A. Raman, Resonant anti-reflection metasurfaces for infrared transmission optics, *Nano Lett.* 23 (2023) 8940–8946.
- [19] P. Anusuyadevi, R. Shanker, Y. Cui, A.V. Riazanova, M. Jarn, M. Jonsson, A. Svagan, Photoresponsive and polarization-sensitive structural colors from cellulose/liquid crystal nanophotonic structures, *Adv. Mater.* 33 (36) (2021) e2101519.
- [20] D. Chae, M. Kim, P. Jung, S. Son, J. Seo, Y. Liu, B. Lee, H. Lee, Spectrally selective inorganic-based multilayer emitter for daytime radiative cooling, *ACS Appl. Mater. Interfaces* 12 (7) (2020) 8073–8081.
- [21] K. Ly, X. Liu, X. Song, C. Xiao, P. Wang, H. Zhou, T. Fan, A dual-mode infrared asymmetric photonic structure for all-season passive radiative cooling and heating, *Adv. Funct. Mater.* 32 (31) (2022) 2203789.
- [22] J. Lee, D. Im, S. Sung, J. Yu, H. Kim, J. Lee, Y. Yoo, Scalable and efficient radiative cooling coatings using uniform-hollow silica spheres, *Appl. Therm. Eng.* 254 (2024) 123810.
- [23] D. Hong, Y. Lee, O. Jeon, I. Lee, S. Lee, J. Won, Y. Jeon, Y. La, S. Kim, G. Park, Y. Yoo, S. Park, Humidity-tolerant porous polymer coating for passive daytime radiative cooling, *Nat. Commun.* 15 (1) (2024) 4457.
- [24] Y. Chen, B. Dang, J. Fu, C. Wang, C. Li, Q. Sun, H. Li, Cellulose-based hybrid structural material for radiative cooling, *Nano Lett.* 21 (1) (2021) 397–404.
- [25] J. He, Q. Zhang, Y. Wu, Y. Ju, Y. Wang, S. Tang, Scalable nanofibrous silk fibroin textile with excellent mie scattering and high sweat evaporation ability for highly efficient passive personal thermal management, *Chem. Eng. J.* 466 (2023) 143127.
- [26] C. Deng, Q. Mo, Z. Hu, D. Yang, Y. Li, T. Zhang, F. Qiu, Eco-friendly and large-scale fabrication of cellulose-based membrane with wrinkled structure for efficient daytime radiative cooling, *Chem. Eng. J.* 512 (2025) 162035.
- [27] Y. Chen, Y. Dai, Y. Zhu, S. Xue, F. Qiu, T. Zhang, Transparent cellulose acetate/polyvinylidene difluoride films with heat dissipation for agricultural mulch application, *Ind. Crops Prod.* 224 (2025) 120302.
- [28] Y. Chen, Y. Chen, Y. Dai, D. Yang, F. Qiu, T. Zhang, Cellulose nanofibers based composite membrane with high solar radiation and heat conduction for agricultural thermal dissipation application, *Sol. Energy* 267 (2024) 112242.
- [29] E. Pakdel, M. Naebe, L. Sun, X. Wang, Advanced functional fibrous materials for enhanced thermoregulating performance, *ACS Appl. Mater. Interfaces* 11 (14) (2019) 13039–13057.
- [30] X. Wu, Y. Wang, X. Liang, Y. Zhang, P. Bi, M. Zhang, S. Li, H. Liang, S. Wang, H. Wang, H. Lu, Y. Zhang, Durable radiative cooling multilayer silk textile with excellent comprehensive performance, *Adv. Funct. Mater.* 34 (2023) 2313539.
- [31] Y. Du, W. Wang, J. Mei, L. Zhang, Silica-bridged inorganic-organic hybrid membrane for efficient daytime radiative cooling, *Chem. Eng. J.* 485 (2024) 149976.
- [32] T. Li, H. Sun, M. Yang, C. Zhang, S. Lv, B. Li, L. Chen, D. Sun, All-ceramic, compressible and scalable nanofibrous aerogels for subambient daytime radiative cooling, *Chem. Eng. J.* 452 (2023) 139518.

- [33] J. Mandal, Y. Fu, A.C. Overvig, M. Jia, K. Sun, N.N. Shi, H. Zhou, X. Xiao, N. Yu, Y. Yang, Hierarchically porous polymer coatings for highly efficient passive daytime radiative cooling, *Science* 362 (6412) (2018) 315–319.
- [34] X. Wang, X. Liu, Z. Li, H. Zhang, Z. Yang, H. Zhou, T. Fan, Scalable flexible hybrid membranes with photonic structures for daytime radiative cooling, *Adv. Funct. Mater.* 30 (5) (2019) 1907562.
- [35] D. Li, X. Liu, W. Li, Z. Lin, B. Zhu, Z. Li, J. Li, B. Li, S. Fan, J. Xie, J. Zhu, Scalable and hierarchically designed polymer film as a selective thermal emitter for high-performance all-day radiative cooling, *Nat. Nanotechnol.* 16 (2) (2020) 153–158.
- [36] Y. Han, Z. Meng, Y. Wu, S. Zhang, S. Wu, Structural colored fabrics with brilliant colors, low angle dependence, and high color fastness based on the mie scattering of Cu_2O spheres, *ACS Appl. Mater. Interfaces* 13 (48) (2021) 57796–57802.
- [37] C. Wu, H. Geng, S. Tan, J. Lv, H. Wang, Z. He, J. Wang, Highly efficient solar anti-icing/deicing via a hierarchical structured surface, *Mater. Horiz.* 7 (8) (2020) 2097–2104.
- [38] K. Lin, S. Chen, Y. Zeng, T. Ho, Y. Zhu, X. Wang, F. Liu, B. Huang, C. Chao, Z. Wang, C. Tso, Hierarchically structured passive radiative cooling ceramic with high solar reflectivity, *Science* 382 (6671) (2023) 691–697.
- [39] J. Yun, D. Chae, S. So, H. Lim, J. Noh, J. Park, N. Kim, C. Park, H. Lee, J. Rho, Optimally designed multimaterial microparticle–polymer composite paints for passive daytime radiative cooling, *ACS Photonics* 10 (8) (2023) 2608–2617.
- [40] C. Zhou, R. Chu, R. Wu, Q. Wu, Electrospun polyethylene oxide/cellulose nanocrystal composite nanofibrous mats with homogeneous and heterogeneous microstructures, *Biomacromolecules* 12 (7) (2011) 2617, 25.